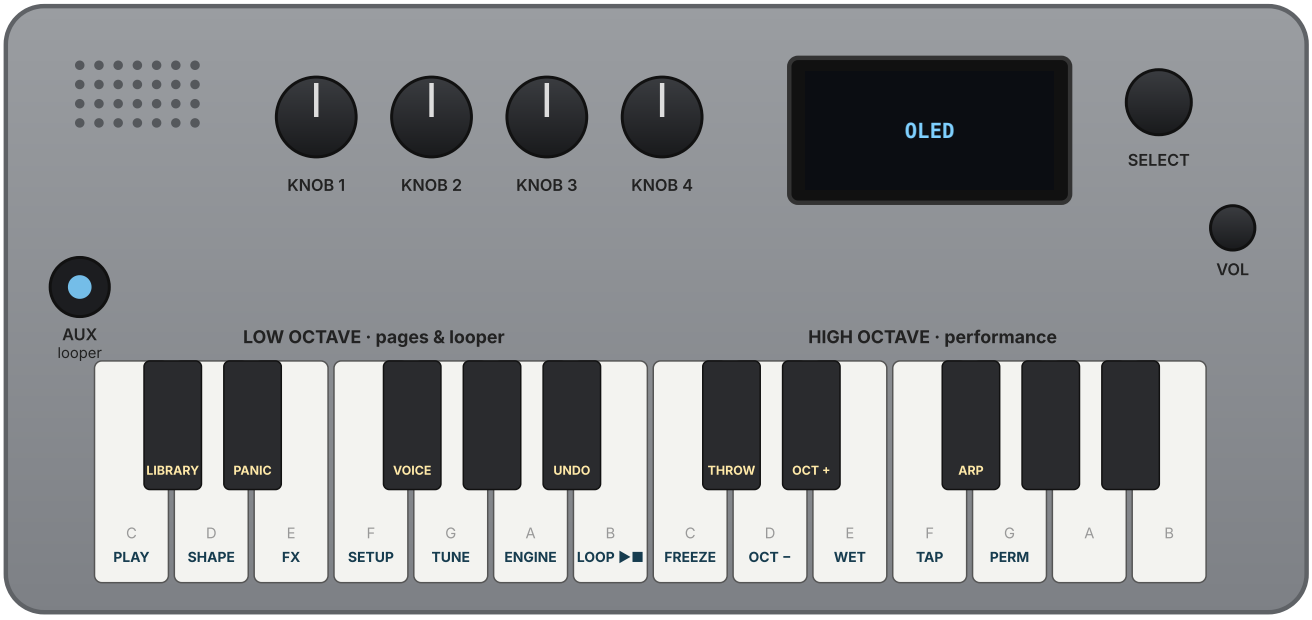
DM48 Engine

A breath-controlled MPE synth host, MIDI looper, arpeggiator and DM48X tuning library for the Critter & Guitari Organelle M — made for the Lekholm DM48X digital harmonica.



Organelle OS 5 · Pure Data vanilla (no externals) · Patch folder: **DM48 Engine**
The illustration is stylised; key labels show what each Organelle key does in this patch.

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1. Quick start

1. Copy the **DM48 Engine** folder into your Patches folder (USB drive, SD card, or the OS 5 web editor).
2. Plug the DM48X into the Organelle's USB port. Go to **System › MIDI Setup › Device**, choose the DM48X, and **Save**.
3. Set up the DM48X (Global settings):

DM48X setting	Value	Why
MIDI channel	**Multich. 2-13**	One channel per hole (MPE). Channel 1 stays free for global messages.
Mono/poly	**Poly**	Chords and the arpeggiator need more than one note.
Double notes	**Allow**	Both C holes in Solo tuning can sound.
Breath CC#	**11** (default) or **Aftertouch**	Breath is read from CC 2, 7, 11, channel pressure or poly pressure.
Breath sens	anything but **Off**	Off sends no breath; notes then only use their start velocity.
Bend range	see §11	Match the patch's *Note Bend* setting.

4. Load **DM48 Engine**. Play. Tap **Aux** (or the footswitch) to start a loop.

> If the DM48X is plugged in after the patch is loaded, reload the patch so the Organelle connects to it.

2. How it is organised

The DM48X is the instrument; the Organelle is the sound engine and the control surface. Every note keeps its own MIDI channel, so breath, bends and timbre shape each note on its own (MPE). Signal flow:

```
DM48X → MIDI in → arpeggiator → looper (records) → voices (8, any engine) → delay →
reverb → limiter → out
                                          |                                          └►
MIDI out echo + clock
                                          └─ loop playback on its own channels (no clashes
with live notes)
```

Breath does far more than volume: each engine maps it to several small timbre changes (brightness, harmonics, noise, detune, drive). **Breath>Tone** on the SHAPE page scales all of them at once.

3. The keyboard as a control surface

You play the DM48X, so the Organelle's 24 keys are free for control. The low octave selects pages and runs the looper; the high octave holds performance effects.



Key	Low octave	High octave
C	PLAY page	Reverb **freeze** on/off
C♯	LIBRARY page	Delay **throw** (while held)
D	SHAPE page	Octave down on/off
D♯	**Panic**: silence everything	Octave up on/off
E	FX page	**Wet only** (dry signal off) on/off
F	SETUP page	**Tap tempo** — two taps less than 2 s apart
F♯	VOICE page	**Arpeggiator** on/off
G	TUNE page	Arp **permutations** on/off
G♯	—	Screen **view**: pages › scope › audio loop waveform
A	ENGINE page	**Audio loop**: record › play › overdub (hold 1 s: clear)
A♯	Loop **undo** last overdub	Audio loop **undo** (newest layer)
B	Loop **stop / start**	Audio loop **stop / start**

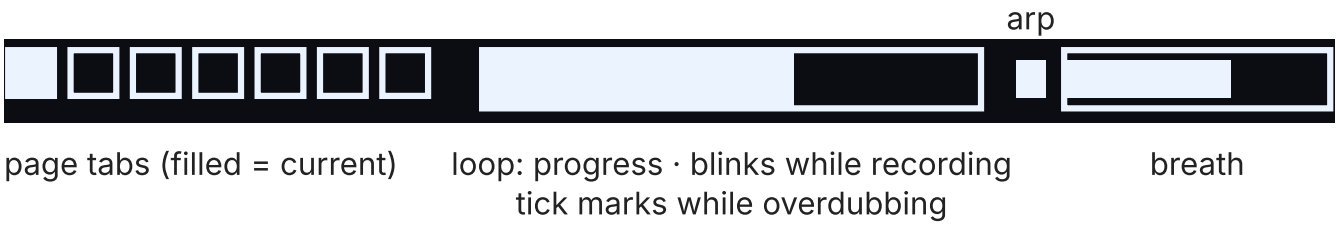
Keys act when you **release** them. If you turn a knob while holding a key, the knob does a **combo** instead and the key's normal action is skipped:

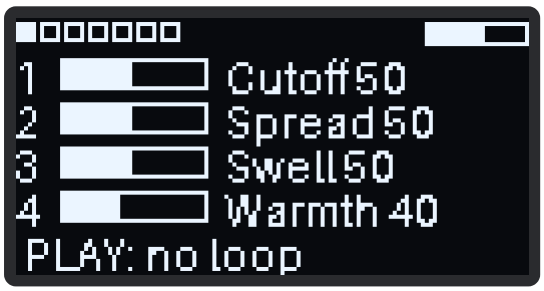
Hold	Turn	Result
B (loop)	any knob	**Loop speed and direction.** Centre = normal. Right = forward, faster the further you turn (up to 4×). Left = reverse, faster the further you turn (up to 4×). The MIDI clock follows.
C (freeze)	any knob	Reverb size
C♯ (throw)	any knob	Throw feedback (how long the throw rings; above 100 it slowly grows)
E (wet)	any knob	Reverb amount
F (tap)	any knob	Tempo 40–200 BPM
F♯ (arp)	knob 1 / 2 / 3	Arp rate / pattern / octave range
A (high, audio loop)	knob 1 / 2 / 3 / 4	Audio loop **speed & direction** (tape style, continuous) / **pitch** −12…+12 semitones, continuous (not in steps; a small dead zone at the centre is exactly 0) / **level** / **layer fade**
B (high, audio loop)	any knob	Record **Pre** (left) or **Post** (right) effects

Aux button and **footswitch** run the looper (§7). On the LIBRARY page, Aux sends a tuning instead; the footswitch still runs the looper.

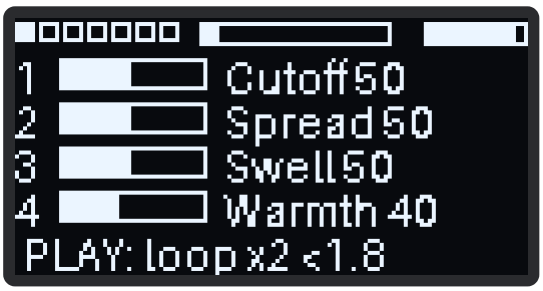
4. The screen

Rows 1–4 show the four knobs of the current page in the PLAY-patch style: knob number, a bar for its value, then its name and value. Rows whose value is a word (scale, key, engine, MIDI mode) show plain text. Line 5 shows the page name and the looper state, and briefly shows messages when you press a key or turn a combo. The top strip is always visible:

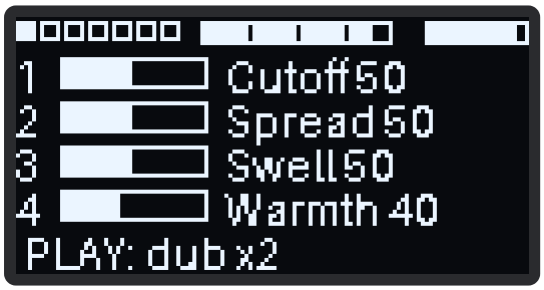




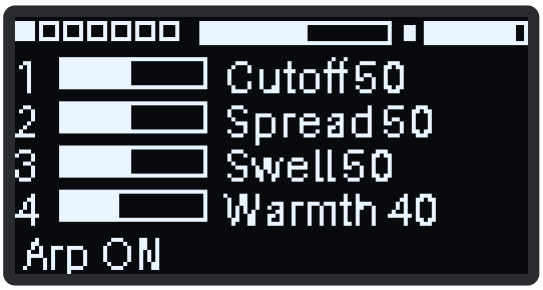
PLAY page with the Velvet engine. Strip: page 1 of 8, breath meter on the right.



Loop playing in reverse at 1.8× (<1.8). The loop bar shows the position.



Overdubbing: tick marks on the loop bar.



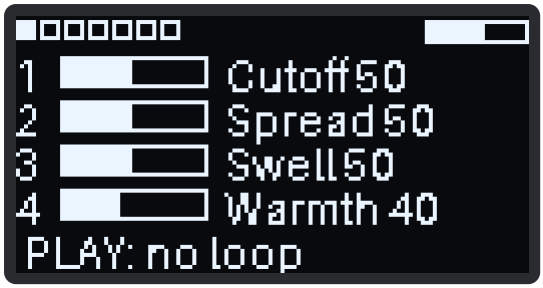
Arpeggiator on: dot next to the breath meter.

5. Pages and knobs

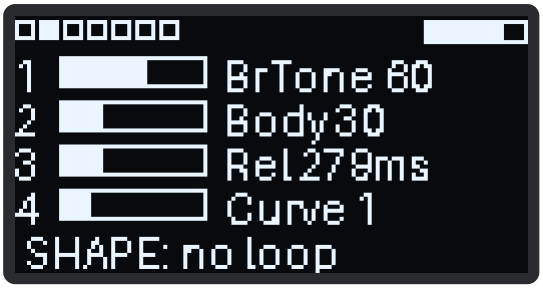
Switch pages with the low-octave keys. A knob changes its setting only once you move it. The smooth settings (PLAY, SHAPE, FX, Level) pick up where they are: turning the knob moves the value from its current setting and it meets the knob at the end of its travel, so nothing jumps when you change page. Selector knobs (scales, engine, octave…) go straight to the knob's position.

PLAY — the engine

The four knobs belong to the loaded engine; their names change with it (see §6). Each engine remembers its own settings.



PLAY (Velvet)



SHAPE

SHAPE — breath and envelope

Knob	Control	What it does
1	**Breath>Tone**	How much breath changes the timbre (all engines). 0 = breath only changes volume.
2	**Body**	A sine one octave below every note.
3	**Release**	Fade after a note ends (10 ms – 3 s).
4	**Breath Curve**	0.5 = easy to get loud · 1 = linear · 3 = needs strong breath.

FX

Knob	Control	What it does
1	**Delay**	20 – 1500 ms. Tap tempo sets it too.
2	**Feedback**	Repeats.
3	**Delay Mix**	Delay level.
4	**Reverb**	Reverb level.

SETUP

Knob	Control	What it does
1	**Octave**	−2 … +2 octaves (the high-octave D / D♯ keys add another ±1).
2	**Note Bend**	Bend range for channels 2–16: 1, 2, 3, 12, 24 or 48 semitones (MPE standard is 48).
3	**Ch1 Bend**	Bend range on channel 1, which bends every note.
4	**Vibrato**	Vibrato speed. Depth comes from CC 1 (DM48X breath modulation or ribbon).

TUNE

Knob	Control	What it does
1	**Family**	Western · Temperament · Gamelan · African · Japanese · Chinese · SE-Asian · Maqam · Indian · Byzantine.
2	**Scale**	A scale within the family (78 in all — see the appendix). Line 4 shows its number in the family.
3	**Key**	The note the scale is counted from: set it to the key of your harp tuning.

Western includes **Just 5-limit** (9/8 5/4 4/3 3/2 5/3 15/8 …) and **Just 7** (the 4th is 21/16 and the minor 3rd / flat 7th are 7/6 / 7/4: on a C harp the draw chord G-B-D-F becomes the pure 4:5:6:7 chord).

ENGINE

Knob	Control	What it does
1	**Engine**	Choose the sound engine (all .pd files in engines/).
2	**MIDI**	MPE (each channel is its own note) or Standard (all channels merged into one).
3	**Level**	Output level.
4	**MIDI out**	Off · Echo · Echo+Clock. Line 4 shows the MIDI arriving from the DM48X.

VOICE — legato and glide

Knob	Control	What it does
1	**Mode**	**Poly** (a voice per note) or **Mono legato**: one voice; a new note while one is held moves the voice there without a new attack, and releasing it goes back to the note still held.
2	**Glide**	Portamento time, 0 (off) – 2 s. Bends during a glide don't cut it short.
3	**When**	**Glide always**, or **Glide legato**: only from notes you are still holding (overlap the notes to slide).
4	**From**	**From nearest**: polyphonic portamento — each new note slides from the nearest note of the chord before (recent chords first), so a chord slides into the next chord. **From last**: every new note slides from the last note played.

Glide works with every engine: the engine receives the start pitch, then a ramp to the new pitch. In Mono legato the engine gets no new note at all, only the ramp, so strings keep ringing and envelopes don't restart.



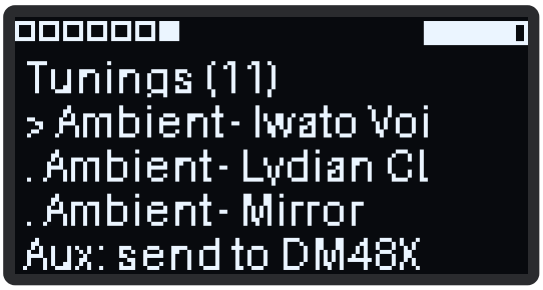
FX



TUNE: family, scale, key



ENGINE: line 4 shows incoming MIDI



LIBRARY

6. Engines

An engine is one voice; the patch runs eight. All fourteen bundled engines react to breath in several small ways at once, and they are level-matched, so switching doesn't jump in volume.

Switching engines while you play. Every engine is loaded once when the patch starts and stays switched off (costing nothing) until you choose it. Changing the engine fades the old one out and the new one in within about 30 ms, and the notes you are holding carry on with the new sound. Nothing is loaded at that moment, so there is no pop. Turning the engine knob quickly through several engines switches once, to where the knob stops.

Engine	Character	Knobs	What breath does
Feral	raw, unstable, weird	Sync · Snarl · Tone · Tail	The most breath-hungry engine, built from scratch. A capacitor-charge oscillator bends from a smooth ramp into a charging curve as you blow, and a hard-synced second oscillator screams as breath sweeps its ratio (Sync sets the range). Everything goes through a vactrol low-pass gate: breath opens volume and brightness together, instantly, and it closes with a slow tail (Tail). Fast changes in breath — puffs and accents — jab the pitch sharp, kick the sync, add crackle and drive the asymmetric diode clipper (Snarl) harder. Soft breath is very quiet and dark; full breath is loud and bright, with a wide range in between.
Volt	raw, electric	Fold · Grit · Detune · Tone	More wave-folding, more sample-rate grit, a brighter filter and a slight upward strain of pitch.
Halo	ethereal, resonant	Shimmer · Resonance · Air · Bloom	Upper partials bloom in (Bloom sets how fast), breath noise rings two tuned resonators, which brighten.
Under	bass, subharmonic	Subs · Drive · Tone · Octave	Deeper subharmonics (f/3, f/4) fade in, more saturation, the filter opens. Octave knob: 0, −1, −2.
Velvet	lush analog pad	Cutoff · Spread · Swell · Warmth	The filter opens, the three saws spread wider, warmth and a slow pitch drift deepen. Swell softens the attack.
Steel	pedal steel string	Tone · Sustain · Steel · Attack	A waveguide string bowed by breath (no pluck): breath is the volume pedal and the bow, brightens the string and adds a little drive. Attack sets how fast the sound swells in when you blow, from 5 ms (instant) to 2 s (a slow volume-pedal swell). Steel adds the stiff, metallic stretch of the partials.
Lyra	LIRA-8 style drone	Mod · Sharp · Detune · Drive	Two triangle oscillators modulate each other. Breath deepens the cross-modulation, sharpens the waves toward squares, widens the beating and pushes the drive. Pitch stays steady.
Hybrid	source × body, 16 combinations	Source · Body · Color · Shape	**Source** (knob value 0–24 Phase, 25–49 Vowel, 50–74 Pulse, 75–100 Breath): CZ-style phase distortion with a resonant wave (breath is the brightness envelope); a formant voice that breath and Color move through u-o-a-e-i; a Juno-style pulse with PWM (breath speeds and widens it) and a sub an octave down; or pitched breath noise with a flute tone. **Body** (same ranges: Filter, Modal, Strings, Tube): a resonant low-pass that breath opens; a 4-mode resonator that Shape bends from string to bell; three sympathetic strings at f, 2f and 3f that ring along (Shape = sustain); or a tube-like saturation that breath drives harder. Color shapes the source, Shape the body. Only the chosen source and body run; turning either knob crossfades in 10 ms.

Doom	clean drone into a dirty chain	Chain · Mod · Dirt · Tone	Lyra-style cross-modulating triangles (Mod) feed one dirty chain (Chain: 0–24 Rip, 25–49 Tear, 50–74 Metal, 75–100 Flesh): Rip is hard drive into a bitcrusher and sample-rate reduction; Tear a swept 4-stage phaser; Metal inharmonic ring modulation into a dispersive spring-like echo; Flesh a saturating feedback loop. Breath pushes the sound from clean to dirty (Dirt sets how far) and ducks the clean side. Tone is the final low-pass.
Frasil	ambient drone (after FRASIL)	Layers · Souffle · Motion · Timbre	Three layers: a choir of three detuned saws (Layers left), additive partials (Layers right) and *souffle*, breath noise ringing resonators tuned to the note (Souffle). Every voice drifts slowly on its own (Motion), and a slow tide moves the brightness. Breath swells the voice and opens it: the choir's filter opens and the upper partials come in. MPE timbre (CC 74) opens it per note. Each layer only runs (costs CPU) while it is turned up.
Glint	icy FM bells (after FRASIL's GLINT)	Ratio · Bright · Bloom · Ping	A two-operator FM voice. Breath sets the level and opens the FM index: soft breath is nearly a pure tone, harder breath brings in the bell's sidebands (Bloom: how much). Ratio picks the modulator (FRASIL's 13 ratios, ×0.5 to ×13.6; ×3 by default). Ping adds a struck accent at the start of a note (0 = none, purely breath). A vibrato fades in as a note settles. MPE timbre brightens each note.
Metal	sung metal bodies (after Jaffedrone)	Body · Bloom · Shimmer · Tone	A resonant body from Jaffedrone's mode tables, picked with Body (in eighths: gamelan gender, bonang, gong, singing bowl, handpan, steelpan, glass, the golden-ratio "alien"). Played by breath, not struck: soft breath sounds the low modes, harder breath brings the upper ones in (Bloom: how far). The fundamental beats against a twin (Shimmer), and breath noise rubs the body.
String	bowed string (after Jaffedrone)	Bright · Buzz · Stiff · Sustain	A string kept ringing by a noise bow, with a jawari bridge (Buzz: a loud note buzzes, like a tanpura) and stiffness (Stiff: the partials stretch, bell-like). Breath is the bow: harder breath is louder and brighter, and buzzes more.
Wind	reeds (after Jaffedrone)	Reed · Bright · Air · Vibrato	A bore driven at the mouthpiece. Reed picks (in thirds) a single reed (clarinet-like, odd harmonics), a double reed (all harmonics) or a free reed (harmonium). Breath is the mouth pressure: soft breath is mellow, harder breath louder and brighter. Air is breath noise, Vibrato a player's vibrato.

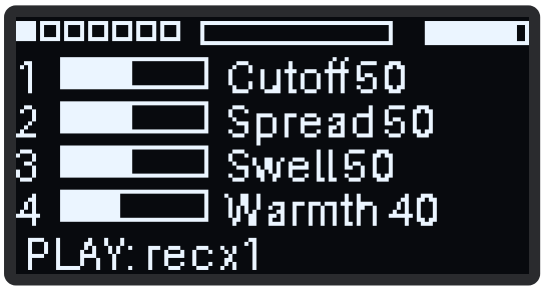
Steel is based on **Steelie** by gur_shulman (WTFPL). Lyra is based on **LIRA-8** by Miguel Moreno (BSD licence; after the Soma Laboratory Lyra-8). Frasil and Glint are after **FRASIL** by Styves Goupil (MPL-2.0; reimplemented in Pd from its design, no code copied). Metal, String and Wind are ported from **Jaffedrone** by mercurious and sydilaxe (MIT). Doom's chains follow ideas from **DOOM_SYNTH**, and Hybrid's sources and bodies ideas from Casio CZ, Braids, Elements and Rings; all are reimplementations, no code was copied.

CC 74 (MPE timbre, if your controller sends it) also brightens each engine per note.

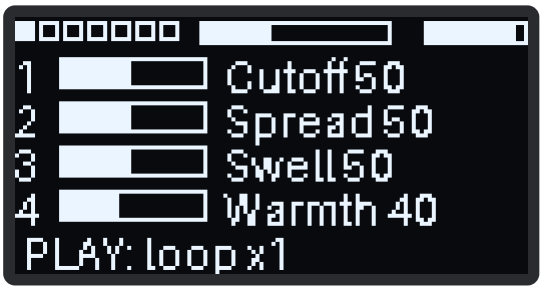
7. Looper

The looper records **MIDI**, not audio: notes, per-note bends, breath and timbre, on their own channels. Playback goes through the current engine, so you can change engine, tuning or effects while a loop plays.

Action	How
Record	Tap Aux (or footswitch). Line 5: `rec`, loop bar blinks.
Close the loop and play	Tap again. The loop length is the time you recorded.
Overdub	Tap while playing (`dub x2`); tap again to keep playing. Up to 7 layers.
Undo last layer	Low A♯.
Stop / start	Low B.
Clear	Hold Aux (or footswitch) for 0.7 s.
Speed and direction	Hold low B and turn any knob: centre normal, right forward faster, left reverse faster.



Recording



Playing



Turning the speed combo

In reverse, notes still begin with their note-on: the looper swaps note-ons and note-offs when it runs backwards. Each loop layer plays on its own set of channels, so loop notes never cut off live notes or each other.

Audio looper

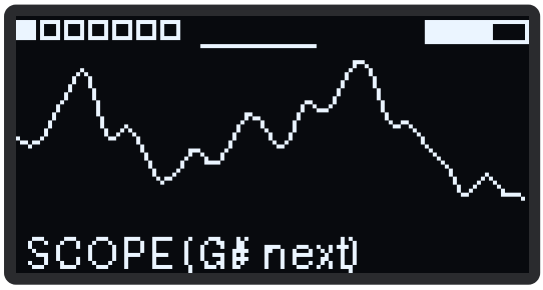
The MIDI loop always plays through the current engine. The **audio looper** records the sound itself, so you can layer different engines: record a pass with Halo, switch to Steel, overdub, and so on. Up to 6 stereo layers, 45 s long.

Action	How
Record / play / overdub	Tap high A: record › play › overdub › play … (a thin line under the loop bar shows the audio loop)
In time with a MIDI loop	If a MIDI loop is playing, the audio loop waits for its start and records exactly one cycle, so both stay in time.
Undo	High A♯ removes the newest layer (again for the one before).
Stop / start	High B.
Clear	Hold high A for 1 s.
Speed & direction	Hold high A + knob 1: varispeed like tape, continuous, pitch follows speed. Centre = 1×; right speeds up to 4×; left slows down (0.5× is an octave down) to a near-stop about a third of the way, then plays backwards, up to −1× at full left.
Pitch	Hold high A + knob 2: −12…+12 semitones, speed unchanged.
Level / layer fade	Hold high A + knob 3 / knob 4. Layer fade makes older layers fade a little each pass (off at the right end).
Pre / Post	Hold high B + a knob. **Pre** records the synth before the effects; the loop then goes through the delay and reverb you set now. **Post** records the finished sound with its effects.

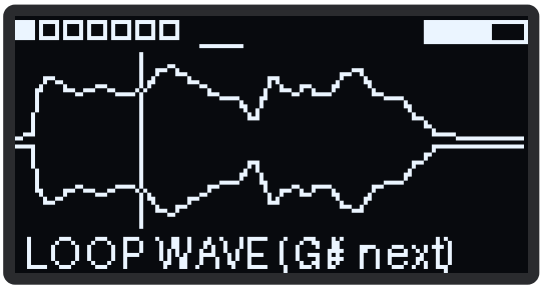
Overdub at any speed, like tape. Each layer remembers the speed (and direction) it was recorded at. Overdub at 2× and then bring the loop back to 1×: that layer plays at half speed, an octave lower and stretched, while the first take plays normally. A layer recorded in reverse plays backwards when the loop runs forwards. Changing the speed ends an overdub; the next one is a new layer at the new speed. A layer holds up to 45 s, so a long loop can't be overdubbed very slowly (the screen says so). Each layer records for at least one whole pass, so undo always removes exactly what you played.

Screen views

High G♯ switches the screen between the pages, a **scope** of the output (steady on the note's waveform, auto-scaled), and the **audio loop waveform** with its playhead. Any page key returns to the pages. The top strip and line 5 stay.



Scope (high G#)



Audio loop waveform with playhead

8. Arpeggiator

High F♯ turns it on. Hold notes on the DM48X and the arpeggiator plays them one at a time — each on its own channel, so breath and bends still shape every note. It runs at the tap tempo.

Hold F♯ +	Sets
knob 1	Rate: 1/4, 1/8, 1/8 triplet, 1/16, 1/16 triplet, 1/32
knob 2	Pattern: up, down, up-down, pedal (lowest note alternating), random
knob 3	Octave range 1–3

Breath shapes the notes, not the pattern:

- **Density** — soft breath plays only some steps; stronger breath fills them in; near full breath some steps double (ratchets).
- **Scatter** — above half breath, notes jump an octave up or down (two octaves when strong) and drift off the grid.

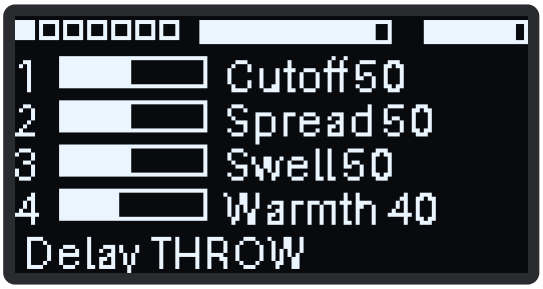
Permutations (high G, on by default): each pass through the pattern rotates the note order, and every other pass is mirrored. The looper records what the arpeggiator plays.

9. Effects

Master chain: stereo ping-pong **delay** (warm-filtered repeats) → **reverb** (an 8-line feedback delay network after the VerbRack patch: four diffusers, slowly moving delay lines for a smooth tail, highs die away faster than lows; size sets 1.8–15 s of decay; mono-safe for the built-in speaker) → **limiter** (always on, with a soft clipper after it). **Freeze** holds the reverb tail indefinitely and mutes its input. **Throw** pushes the delay into long feedback while you hold the key. **Wet only** removes the dry sound for washes. **Tap tempo** sets the delay time, the arp and the MIDI clock.



Freeze toggled on



Delay throw

10. Tuning: just intonation and the DM48X tuning library

Just intonation in the synth

MIDI notes are equal-tempered, so the tuning happens in the patch: every note is retuned on its own, so chords stay pure in any mix. The TUNE page offers 78 scales in ten families — just intonation, historical temperaments, gamelan slendro and pelog, African xylophone and kora tunings, Japanese, Chinese and Southeast Asian scales, maqam and quarter-tone scales, Indian ragas and Byzantine modes. Each is a set of 12 offsets in cents, counted from *Key*. The world-music scales come from the **Astral** patch's scale library. Pitches are measured exact in testing (e.g. E in Pelog Degung Sejati is +24 cents).

The LIBRARY page

Sends tunings (MIDI files in the `tunings/` folder) to the DM48X, without stopping the loop or touching the sound.

1. On the DM48X: **Tuning editor** › step past the last hole › **Receive tuning**.
2. On the Organelle: low C♯ (LIBRARY). Knob 2 picks a folder — *All*, then each subfolder of `tunings/` (Ambient, Chromatic, Frasil, Richter, World); line 1 shows the folder and how many tunings it holds. Knob 1 picks the tuning, **Aux** sends it. Line 5 says *Sent!*.
3. On the DM48X: hold the round button for 2 s on the Tuning page to save it to a slot.

> First time: send **Order test** and check in the Tuning editor that hole 1 blow = C3 (MIDI 48), hole 1 draw = C4 (60), hole 1 blow + slider = C5 (72) and hole 12 draw + slider = B6 (95). The 48-note order was read from the DM48X firmware; this confirms it on your instrument. Nothing is stored until you save on the DM48X.

Key follows the tuning. If the tuning's name ends in a key (*Richter Low D*, ... *Bb*, ... *F#*), sending it also sets the TUNE page Key, so the synth's just intonation is built on the same key as the harp. Line 5 shows *Sent! Synth key D*. You can still change Key afterwards. The DM48X itself only stores ordinary note numbers; the pure tuning happens in the synth.

Frasil tunings (folder *Frasil*) are made for ambient drones: the blow holes walk one chord and the draw holes a second chord that sits well with it (for example Cmaj7 and Dm in *Glacier C*), so any holes you breathe together sound consonant. The plain holes are the **lower octave**; the **slider jumps the same layout up an octave**. Glacier C (Ionian), Lydian Haze D, Dusk A (Dorian), Aeolian Tide F, Snow Pentatonic G, Open Fifths E (quartal), Mixolydian Eb and Harmonics D (the overtone series). Each name ends in its key, so sending it also sets the TUNE Key.

To add your own, record the DM48X's **Send tuning** as a MIDI file and put it in `tunings/` or any folder inside it (a new folder appears on knob 2 the next time the patch loads). Format 0 and 1 files work; only the notes are sent.

> The Organelle has one MIDI output, so tuning notes also reach anything on the hardware MIDI Out jack. MIDI echo pauses while a tuning is sent, and MIDI coming *from* the DM48X is ignored until 0.6 s after it ends, so the instrument's replies can't trigger notes or a pop.

11. MIDI

MPE (default): channels 2–16 are notes (each with its own bend, breath and timbre); a bend on channel 1 bends every note. **Standard**: all channels merge into one, for a DM48X in single-channel mode.

Bend ranges: the DM48X's breath bends and ribbon send pitch bend. Either set *Note Bend* (SETUP knob 2) to the range the DM48X expects (often 2), or keep 48 (MPE standard) and lower the DM48X's *Bend range* so its bends stay small.

Messages read: notes, pitch bend, channel pressure, poly pressure, CC 2 / 7 / 11 (breath), CC 74 (timbre), CC 1 (vibrato depth), CC 120 / 123 (all notes off).

MIDI out (ENGINE knob 4): *Echo* repeats everything the engine plays — live and looped, on the original channels — to the Organelle's MIDI out. *Echo+Clock* adds 24-per-beat MIDI clock at the tap tempo (following the loop speed), Start when a loop starts playing, and Stop when it stops or is cleared. The Organelle's own MIDI out (knob CCs, key notes) is switched off while this patch runs.

> The Organelle's System › MIDI Setup › *MIDI In Ch.* only affects the Organelle's built-in MIDI functions. If the DM48X sends program changes, set it to a channel the DM48X never uses (e.g. 16) so the Organelle doesn't switch patches.

12. Making your own engine

Copy an engine in `engines/`, rename it and change it. It appears on the ENGINE page after a reload. An engine is one voice:

```
inlet 1 (left)  pitch   "note ms"   MIDI note, fractional (bend, tuning, octave included) —
feed it to [line~]
inlet 2         level   "level ms"  0..1 breath/velocity, already curved; 0 with the
release time at note-off
inlet 3         timbre  "value ms"  0..1, 0.5 = neutral (CC 74)
outlet~         mono audio

Shared:  [r~ $1-p1~] .. [r~ $1-p4~]  PLAY knobs (0..1, smoothed)   [r $1-p1] .. control
rate
         (every engine in the folder is created once at start-up and switched off until
selected)
         [r~ eng-bamt~]  Breath>Tone    [r~ eng-body~]  Body    [r~ eng-vib]  vibrato
(semitones)
         [r~ eng-sawk]   for lib/bsaw (band-limited saw)

engines/NAME.txt   line 1: four knob names    line 2: four default values
```

Any Pd synth voice can be wrapped this way: connect pitch to its frequency (via `[mtof~]`), level to its amplitude, and send its output to the outlet. Let breath change several things a little rather than one thing a lot. Keep it light — eight copies run at once.

13. Performance on the Organelle M

The Organelle M runs on a Raspberry Pi Compute Module 3 (four ARM Cortex-A53 cores at 1.2 GHz). Pure Data computes all audio on one of those cores, so that single core is the budget. The patch is built for it:

- Each voice switches its DSP off when silent, so only sounding voices cost anything. The audio looper's DSP is off until you first record; its 6 × 45 s of stereo buffers take about 95 MB of the M's 1 GB, reserved when the patch loads.
- Saws are band-limited without oversampling; filters are cheap 2-pole designs (Velvet cascades two for 4-pole warmth).
- All control logic is message-based; the arpeggiator, looper and screen run at control rate.

The effects, the looper and all engines use only Pd's fast built-in signal objects — no `expr~`, which is slow on the Organelle's ARM processor. Each engine computes its costliest slow signals (pitch, partial levels, filter curves) at an eighth of the sample rate, and the audio looper only runs the layers you have recorded. Measured offline with all 8 voices sounding, the engines use 0.75–1.9 % of one 2.1 GHz Xeon core on top of the effects (Under the lightest, Steel the heaviest; Hybrid 1.0–2.8 % depending on the source and body, Vowel into Modal the heaviest; Doom 1.6–2.0 %; Frasil 2.5–3.0 %, Metal 2.1 %, Wind 1.8–2.1 %, String 1.7 %, Glint 0.6 %), up to 40 % less than before (Halo −41 %, Under −25 %, Lyra −22 %, Feral −22 %, Velvet −15 %); the audio looper adds 0.15–0.3 %, about a third less. A Cortex-A53 core is very roughly 8–10× slower for this kind of work, so expect around 25–

50 % of the Pd core on the Organelle M with eight voices held — less in normal playing. If you hear clicks, play fewer notes at once with Steel or Hybrid's Vowel/Strings, or lower Reverb.

14. Saving, files and troubleshooting

Settings **save themselves** two seconds after you change something (page settings to `settings.txt`, each engine's PLAY knobs to `engines.txt`, in the patch folder). **System › Save Patch** still works too. Loops are not saved. To go back to the defaults, delete `settings.txt`.

```
DM48 Engine/
  main.pd  voice.pd  README.txt  settings.txt  engines.txt
  engines/   feral volt halo under velvet steel lyra hybrid doom frasil glint metal string
wind (.pd + .txt)  lib/bsaw.pd  README.txt
  tunings/   *.mid  TUNINGS.txt
```

Problem	Check
No sound	MIDI Setup › Device is the DM48X; reload the patch after plugging in; ENGINE line 4 shows incoming MIDI.
Breath does nothing	DM48X Breath sens not Off; Breath CC# 11 / 2 / 7 or Aftertouch; the strip's breath meter should move.
Bends far too wide	SETUP Note Bend vs the DM48X Bend range (§11).
Knob does nothing	A key may still count as held — press and release any low-octave key.
A sound keeps going	Press low D♯ (**Panic**): all notes off, both loopers stop (nothing is erased; start them again as usual), freeze, throw and the arpeggiator switch off, and the delay and reverb are emptied. Then please note what you did just before, so it can be fixed.
Pop from the speaker when sending	Should be gone in this version (the DM48X's replies are now ignored while sending). If it persists, it is likely the USB power draw of the DM48X saving — try the headphone/line out, or a powered USB hub.
Tuning didn't arrive	DM48X must be in Receive tuning before you press Aux; check line 5 says Sent!.

> Tested offline in Pure Data 0.54 (the Organelle runs 0.53, and only objects available in 0.53 are used) with recorded test input — not yet on an Organelle with a DM48X. Sounds, levels and CPU will need a round of tuning on the real hardware.

Appendix A: scales (TUNE page)

Family	Scales
Western	Equal 12-TET · Just 5-limit · Just 7 · JI Major 5-limit · JI Minor 5-limit · JI Major 7-limit · Harm Ser 8-16 · Pyth Major
Temperament	1/4comma Mtone maj · 1/4comma Mtone min · 1/4comma Mtone alt · Werck III maj · Werck III min · Werck III alt
Gamelan	Slen Kodok Ngorek · Slen Madoe Kentir · Slen Udan Mas · Pelog Degung Sejati · Pelog Malay · Slen Joged Bumbung · Slen Gender Barung · Slen leg · Pelog leg · Bali Pelog leg
African	Timbila Chopi 1 · Timbila Chopi 2 · Banyoro Xyl · Volans African · Mandinka Kora · Ethiop Fort maj · Ethiop Fort min · Ethiop Fort alt
Japanese	Hirajoshi obs 1 · Hirajoshi obs 2 · Hirajoshi th · Shakuh Honkyoku
Chinese	Chin Sienzi · Chin Dizi · Chin Sheng · Chin Yangqin · Chin Bronze 3c.BC · Chin Zhou Pent · Chin Lu Han maj · Chin Lu Han min · Chin Lu Han alt · Palace Mode
SE-Asian	Dan Tranh Bac · Dan Tranh Vong Co · Burm Hornb · Burm Balafon · Thai Hept · Saung Gauk Harp
Maqam	Zalzal Flute · al-Urmawi Maqam · Huzzam · Segah rat · Turk Palmer · Rast Arab 24-TET · Rast Turk AEU · Bayati · Hijaz pr · 1/4tone 24-EDO maj · 1/4tone 24-EDO min · 1/4tone 24-EDO alt
Indian	Ind Mode A · Ind Mode B · Ind Mode C · Ind Mode E · Raga Todi avg · Raga Malkauns · Raga Gunkali · Bhairav pr · Todi pr · Darbari Kanada · N Ind Gamut maj · N Ind Gamut alt
Byzantine	Chrys Diat · Athan Chrom.

Appendix B: DM48X tunings included

Holes 1–12 left to right; middle C = C4 = MIDI 60. Groups: *Richter Low*, *Country*, *Paddy Richter* and *Harmonic Minor* are diatonic harps in the low octave (slider = +1 octave; Richter Low in all 12 keys); *Chromatic* tunings use the slider for the semitone above, like a chromatic harmonica; *Ambient* tunings are open, chordal layouts; *World* tunings walk a scale (blow, then draw one step up). A name ending in a key sets the TUNE Key when sent: choose Just (or the matching world scale, e.g. Pelog or Slendro) on the TUNE page for the true pitches.

```
Richter / Richter Low G
  blow            D2   G2   B2   D3   G3   B3   D4   G4   B4   D5   G5   B5
  draw            E2   A2   D3  Gb3   A3   C4   E4  Gb4   A4   C5   E5  Gb5
  blow+slider     D3   G3   B3   D4   G4   B4   D5   G5   B5   D6   G6   B6
  draw+slider     E3   A3   D4  Gb4   A4   C5   E5  Gb5   A5   C6   E6  Gb6
```

```
Richter / Richter Low Ab
  blow           Eb2  Ab2   C3  Eb3  Ab3   C4  Eb4  Ab4   C5  Eb5  Ab5   C6
  draw            F2  Bb2  Eb3   G3  Bb3  Db4   F4   G4  Bb4  Db5   F5   G5
  blow+slider    Eb3  Ab3   C4  Eb4  Ab4   C5  Eb5  Ab5   C6  Eb6  Ab6   C7
  draw+slider     F3  Bb3  Eb4   G4  Bb4  Db5   F5   G5  Bb5  Db6   F6   G6
```

```
Richter / Richter Low A
  blow            E2   A2  Db3   E3   A3  Db4   E4   A4  Db5   E5   A5  Db6
  draw           Gb2   B2   E3  Ab3   B3   D4  Gb4  Ab4   B4   D5  Gb5  Ab5
  blow+slider     E3   A3  Db4   E4   A4  Db5   E5   A5  Db6   E6   A6  Db7
  draw+slider    Gb3   B3   E4  Ab4   B4   D5  Gb5  Ab5   B5   D6  Gb6  Ab6
```

```
Richter / Richter Low Bb
  blow            F2  Bb2   D3   F3  Bb3   D4   F4  Bb4   D5   F5  Bb5   D6
  draw            G2   C3   F3   A3   C4  Eb4   G4   A4   C5  Eb5   G5   A5
  blow+slider     F3  Bb3   D4   F4  Bb4   D5   F5  Bb5   D6   F6  Bb6   D7
  draw+slider     G3   C4   F4   A4   C5  Eb5   G5   A5   C6  Eb6   G6   A6
```

```
Richter / Richter Low B
  blow           Gb2   B2  Eb3  Gb3   B3  Eb4  Gb4   B4  Eb5  Gb5   B5  Eb6
  draw           Ab2  Db3  Gb3  Bb3  Db4   E4  Ab4  Bb4  Db5   E5  Ab5  Bb5
  blow+slider    Gb3   B3  Eb4  Gb4   B4  Eb5  Gb5   B5  Eb6  Gb6   B6  Eb7
  draw+slider    Ab3  Db4  Gb4  Bb4  Db5   E5  Ab5  Bb5  Db6   E6  Ab6  Bb6
```

```
Richter / Richter Low C
  blow            G2   C3   E3   G3   C4   E4   G4   C5   E5   G5   C6   E6
  draw            A2   D3   G3   B3   D4   F4   A4   B4   D5   F5   A5   B5
  blow+slider     G3   C4   E4   G4   C5   E5   G5   C6   E6   G6   C7   E7
  draw+slider     A3   D4   G4   B4   D5   F5   A5   B5   D6   F6   A6   B6
```

```
Richter / Richter Low Db
  blow           Ab2  Db3   F3  Ab3  Db4   F4  Ab4  Db5   F5  Ab5  Db6   F6
  draw           Bb2  Eb3  Ab3   C4  Eb4  Gb4  Bb4   C5  Eb5  Gb5  Bb5   C6
  blow+slider    Ab3  Db4   F4  Ab4  Db5   F5  Ab5  Db6   F6  Ab6  Db7   F7
  draw+slider    Bb3  Eb4  Ab4   C5  Eb5  Gb5  Bb5   C6  Eb6  Gb6  Bb6   C7
```

```
Richter / Richter Low D
  blow            A2   D3  Gb3   A3   D4  Gb4   A4   D5  Gb5   A5   D6  Gb6
  draw            B2   E3   A3  Db4   E4   G4   B4  Db5   E5   G5   B5  Db6
  blow+slider     A3   D4  Gb4   A4   D5  Gb5   A5   D6  Gb6   A6   D7  Gb7
  draw+slider     B3   E4   A4  Db5   E5   G5   B5  Db6   E6   G6   B6  Db7
```

```
Richter / Richter Low Eb
  blow           Bb2  Eb3   G3  Bb3  Eb4   G4  Bb4  Eb5   G5  Bb5  Eb6   G6
  draw            C3   F3  Bb3   D4   F4  Ab4   C5   D5   F5  Ab5   C6   D6
  blow+slider    Bb3  Eb4   G4  Bb4  Eb5   G5  Bb5  Eb6   G6  Bb6  Eb7   G7
  draw+slider     C4   F4  Bb4   D5   F5  Ab5   C6   D6   F6  Ab6   C7   D7
```

```
Richter / Richter Low E
  blow            B2   E3  Ab3   B3   E4  Ab4   B4   E5  Ab5   B5   E6  Ab6
  draw           Db3  Gb3   B3  Eb4  Gb4   A4  Db5  Eb5  Gb5   A5  Db6  Eb6
  blow+slider     B3   E4  Ab4   B4   E5  Ab5   B5   E6  Ab6   B6   E7  Ab7
  draw+slider    Db4  Gb4   B4  Eb5  Gb5   A5  Db6  Eb6  Gb6   A6  Db7  Eb7
```

```
Richter / Richter Low F
  blow            C3   F3   A3   C4   F4   A4   C5   F5   A5   C6   F6   A6
  draw            D3   G3   C4   E4   G4  Bb4   D5   E5   G5  Bb5   D6   E6
  blow+slider     C4   F4   A4   C5   F5   A5   C6   F6   A6   C7   F7   A7
  draw+slider     D4   G4   C5   E5   G5  Bb5   D6   E6   G6  Bb6   D7   E7
```

```
Richter / Richter Low F#
  blow           Db3  Gb3  Bb3  Db4  Gb4  Bb4  Db5  Gb5  Bb5  Db6  Gb6  Bb6
  draw           Eb3  Ab3  Db4   F4  Ab4   B4  Eb5   F5  Ab5   B5  Eb6   F6
  blow+slider    Db4  Gb4  Bb4  Db5  Gb5  Bb5  Db6  Gb6  Bb6  Db7  Gb7  Bb7
  draw+slider    Eb4  Ab4  Db5   F5  Ab5   B5  Eb6   F6  Ab6   B6  Eb7   F7
```

```
Richter / Country C
  blow            G2   C3   E3   G3   C4   E4   G4   C5   E5   G5   C6   E6
  draw            A2   D3   G3   B3   D4  Gb4   A4   B4   D5   F5   A5   B5
  blow+slider     G3   C4   E4   G4   C5   E5   G5   C6   E6   G6   C7   E7
  draw+slider     A3   D4   G4   B4   D5  Gb5   A5   B5   D6   F6   A6   B6
```

```
Richter / Country G
  blow            D2   G2   B2   D3   G3   B3   D4   G4   B4   D5   G5   B5
  draw            E2   A2   D3  Gb3   A3  Db4   E4  Gb4   A4   C5   E5  Gb5
  blow+slider     D3   G3   B3   D4   G4   B4   D5   G5   B5   D6   G6   B6
  draw+slider     E3   A3   D4  Gb4   A4  Db5   E5  Gb5   A5   C6   E6  Gb6
```

```
Richter / Country A
  blow            E2   A2  Db3   E3   A3  Db4   E4   A4  Db5   E5   A5  Db6
  draw           Gb2   B2   E3  Ab3   B3  Eb4  Gb4  Ab4   B4   D5  Gb5  Ab5
  blow+slider     E3   A3  Db4   E4   A4  Db5   E5   A5  Db6   E6   A6  Db7
  draw+slider    Gb3   B3   E4  Ab4   B4  Eb5  Gb5  Ab5   B5   D6  Gb6  Ab6
```

```
Richter / Country D
  blow            A2   D3  Gb3   A3   D4  Gb4   A4   D5  Gb5   A5   D6  Gb6
  draw            B2   E3   A3  Db4   E4  Ab4   B4  Db5   E5   G5   B5  Db6
  blow+slider     A3   D4  Gb4   A4   D5  Gb5   A5   D6  Gb6   A6   D7  Gb7
  draw+slider     B3   E4   A4  Db5   E5  Ab5   B5  Db6   E6   G6   B6  Db7
```

```
Richter / Paddy Richter C
  blow            G2   C3   E3   A3   C4   E4   G4   C5   E5   G5   C6   E6
  draw            A2   D3   G3   B3   D4   F4   A4   B4   D5   F5   A5   B5
  blow+slider     G3   C4   E4   A4   C5   E5   G5   C6   E6   G6   C7   E7
  draw+slider     A3   D4   G4   B4   D5   F5   A5   B5   D6   F6   A6   B6
```

```
Richter / Paddy Richter G
  blow            D2   G2   B2   E3   G3   B3   D4   G4   B4   D5   G5   B5
  draw            E2   A2   D3  Gb3   A3   C4   E4  Gb4   A4   C5   E5  Gb5
  blow+slider     D3   G3   B3   E4   G4   B4   D5   G5   B5   D6   G6   B6
  draw+slider     E3   A3   D4  Gb4   A4   C5   E5  Gb5   A5   C6   E6  Gb6
```

```
Richter / Paddy Richter D
  blow            A2   D3  Gb3   B3   D4  Gb4   A4   D5  Gb5   A5   D6  Gb6
  draw            B2   E3   A3  Db4   E4   G4   B4  Db5   E5   G5   B5  Db6
  blow+slider     A3   D4  Gb4   B4   D5  Gb5   A5   D6  Gb6   A6   D7  Gb7
  draw+slider     B3   E4   A4  Db5   E5   G5   B5  Db6   E6   G6   B6  Db7
```

```
Richter / Harmonic Minor A
  blow            E2   A2   C3   E3   A3   C4   E4   A4   C5   E5   A5   C6
  draw            F2   B2   E3  Ab3   B3   D4   F4  Ab4   B4   D5   F5  Ab5
  blow+slider     E3   A3   C4   E4   A4   C5   E5   A5   C6   E6   A6   C7
  draw+slider     F3   B3   E4  Ab4   B4   D5   F5  Ab5   B5   D6   F6  Ab6
```

```
Richter / Harmonic Minor C
  blow            G2   C3  Eb3   G3   C4  Eb4   G4   C5  Eb5   G5   C6  Eb6
  draw           Ab2   D3   G3   B3   D4   F4  Ab4   B4   D5   F5  Ab5   B5
  blow+slider     G3   C4  Eb4   G4   C5  Eb5   G5   C6  Eb6   G6   C7  Eb7
  draw+slider    Ab3   D4   G4   B4   D5   F5  Ab5   B5   D6   F6  Ab6   B6
```

```
Richter / Harmonic Minor D
  blow            A2   D3   F3   A3   D4   F4   A4   D5   F5   A5   D6   F6
  draw           Bb2   E3   A3  Db4   E4   G4  Bb4  Db5   E5   G5  Bb5  Db6
  blow+slider     A3   D4   F4   A4   D5   F5   A5   D6   F6   A6   D7   F7
  draw+slider    Bb3   E4   A4  Db5   E5   G5  Bb5  Db6   E6   G6  Bb6  Db7
```

```
Richter / Harmonic Minor E
  blow            B2   E3   G3   B3   E4   G4   B4   E5   G5   B5   E6   G6
  draw            C3  Gb3   B3  Eb4  Gb4   A4   C5  Eb5  Gb5   A5   C6  Eb6
  blow+slider     B3   E4   G4   B4   E5   G5   B5   E6   G6   B6   E7   G7
  draw+slider     C4  Gb4   B4  Eb5  Gb5   A5   C6  Eb6  Gb6   A6   C7  Eb7
```

```
Chromatic / Solo C
  blow            C3   E3   G3   C4   C4   E4   G4   C5   C5   E5   G5   C6
  draw            D3   F3   A3   B3   D4   F4   A4   B4   D5   F5   A5   B5
  blow+slider    Db3   F3  Ab3  Db4  Db4   F4  Ab4  Db5  Db5   F5  Ab5  Db6
  draw+slider    Eb3  Gb3  Bb3   C4  Eb4  Gb4  Bb4   C5  Eb5  Gb5  Bb5   C6
```

```
Chromatic / Solo G
  blow            G2   B2   D3   G3   G3   B3   D4   G4   G4   B4   D5   G5
  draw            A2   C3   E3  Gb3   A3   C4   E4  Gb4   A4   C5   E5  Gb5
  blow+slider    Ab2   C3  Eb3  Ab3  Ab3   C4  Eb4  Ab4  Ab4   C5  Eb5  Ab5
  draw+slider    Bb2  Db3   F3   G3  Bb3  Db4   F4   G4  Bb4  Db5   F5   G5
```

```
Chromatic / Low C
  blow            C2   E2   G2   C3   C3   E3   G3   C4   C4   E4   G4   C5
  draw            D2   F2   A2   B2   D3   F3   A3   B3   D4   F4   A4   B4
  blow+slider    Db2   F2  Ab2  Db3  Db3   F3  Ab3  Db4  Db4   F4  Ab4  Db5
  draw+slider    Eb2  Gb2  Bb2   C3  Eb3  Gb3  Bb3   C4  Eb4  Gb4  Bb4   C5
```

```
Chromatic / Augmented
  blow            C3   E3  Ab3   C4   E4  Ab4   C5   E5  Ab5   C6   E6  Ab6
  draw            D3  Gb3  Bb3   D4  Gb4  Bb4   D5  Gb5  Bb5   D6  Gb6  Bb6
  blow+slider    Db3   F3   A3  Db4   F4   A4  Db5   F5   A5  Db6   F6   A6
  draw+slider    Eb3   G3   B3  Eb4   G4   B4  Eb5   G5   B5  Eb6   G6   B6
```

```
Chromatic / Diminished
  blow            C3  Eb3  Gb3   A3   C4  Eb4  Gb4   A4   C5  Eb5  Gb5   A5
  draw            D3   F3  Ab3   B3   D4   F4  Ab4   B4   D5   F5  Ab5   B5
  blow+slider    Db3   E3   G3  Bb3  Db4   E4   G4  Bb4  Db5   E5   G5  Bb5
  draw+slider    Eb3  Gb3   A3   C4  Eb4  Gb4   A4   C5  Eb5  Gb5   A5   C6
```

```
Ambient / Overtones
  blow            C2   G2   C3   E3   G3  Bb3   C4   D4   E4  Gb4   G4  Ab4
  draw            G2   E3  Bb3   D4  Gb4  Ab4   B4  Db5  Eb5   F5  Gb5  Ab5
  blow+slider     C3   G3   C4   E4   G4  Bb4   C5   D5   E5  Gb5   G5  Ab5
  draw+slider     G3   E4  Bb4   D5  Gb5  Ab5   B5  Db6  Eb6   F6  Gb6  Ab6
```

```
Ambient / Quartal Ladder
  blow            C2   F2  Bb2  Eb3  Ab3  Db4  Gb4   B4   E5   A5   D6   G6
  draw            D2   G2   C3   F3  Bb3  Eb4  Ab4  Db5  Gb5   B5   E6   A6
  blow+slider     G2   C3   F3  Bb3  Eb4  Ab4  Db5  Gb5   B5   E6   A6   D7
  draw+slider     A2   D3   G3   C4   F4  Bb4  Eb5  Ab5  Db6  Gb6   B6   E7
```

```
Ambient / Lydian Clusters
  blow            C3   D3   E3  Gb3   G3   A3   B3   C4   D4   E4  Gb4   G4
  draw            E3  Gb3   G3   A3   B3   C4   D4   E4  Gb4   G4   A4   B4
  blow+slider     C4   D4   E4  Gb4   G4   A4   B4   C5   D5   E5  Gb5   G5
  draw+slider     E4  Gb4   G4   A4   B4   C5   D5   E5  Gb5   G5   A5   B5
```

```
Ambient / Mystic Tritone
  blow            C3   D3   E3  Gb3   A3  Bb3   C4   D4   E4  Gb4   A4  Bb4
  draw           Gb3  Ab3  Bb3   C4  Eb4   E4  Gb4  Ab4  Bb4   C5  Eb5   E5
  blow+slider     C4   D4   E4  Gb4   A4  Bb4   C5   D5   E5  Gb5   A5  Bb5
  draw+slider    Gb4  Ab4  Bb4   C5  Eb5   E5  Gb5  Ab5  Bb5   C6  Eb6   E6
```

```
Ambient / Iwato Void
  blow            C2  Db2   F2  Gb2  Bb2   C3  Db3   F3  Gb3  Bb3   C4  Db4
  draw            G2  Ab2   C3  Db3   F3   G3  Ab3   C4  Db4   F4   G4  Ab4
  blow+slider     C3  Db3   F3  Gb3  Bb3   C4  Db4   F4  Gb4  Bb4   C5  Db5
  draw+slider     G3  Ab3   C4  Db4   F4   G4  Ab4   C5  Db5   F5   G5  Ab5
```

```
Ambient / Mirror
  blow            D4   E4   F4   G4   A4  Bb4   C5   D5   E5   F5   G5   A5
  draw            D4   C4   B3   A3   G3  Gb3   E3   D3   C3   B2   A2   G2
  blow+slider     D5   E5   F5   G5   A5  Bb5   C6   D6   E6   F6   G6   A6
  draw+slider     D3   C3   B2   A2   G2  Gb2   E2   D2   C2   B1   A1   G1
```

```
Ambient / Open Fifths
  blow            C2   D2   E2   G2   A2   C3   D3   E3   G3   A3   C4   D4
  draw            G2   A2   B2   D3   E3   G3   A3   B3   D4   E4   G4   A4
  blow+slider     C3   D3   E3   G3   A3   C4   D4   E4   G4   A4   C5   D5
  draw+slider     G3   A3   B3   D4   E4   G4   A4   B4   D5   E5   G5   A5
```

```
Ambient / Dorian Thirds
  blow            D3   F3   A3   C4   E4   G4   B4   D5   F5   A5   C6   E6
  draw            E3   G3   B3   D4   F4   A4   C5   E5   G5   B5   D6   F6
  blow+slider     D2   F2   A2   C3   E3   G3   B3   D4   F4   A4   C5   E5
  draw+slider     E2   G2   B2   D3   F3   A3   C4   E4   G4   B4   D5   F5
```

```
Ambient / Pentatonic Rain
  blow            C3   D3   E3   G3   A3   C4   D4   E4   G4   A4   C5   D5
  draw            A3   C4   D4   E4   G4   A4   C5   D5   E5   G5   A5   C6
  blow+slider     C4   D4   E4   G4   A4   C5   D5   E5   G5   A5   C6   D6
  draw+slider     A4   C5   D5   E5   G5   A5   C6   D6   E6   G6   A6   C7
```

```
Ambient / Hexatonic Glass
  blow            C3  Db3   E3   F3  Ab3   A3   C4  Db4   E4   F4  Ab4   A4
  draw            E3   G3  Ab3   B3   C4  Eb4   E4   G4  Ab4   B4   C5  Eb5
  blow+slider     C4  Db4   E4   F4  Ab4   A4   C5  Db5   E5   F5  Ab5   A5
  draw+slider     E4   G4  Ab4   B4   C5  Eb5   E5   G5  Ab5   B5   C6  Eb6
```

```
World / Hijaz D
  blow            D3  Eb3  Gb3   G3   A3  Bb3   C4   D4  Eb4  Gb4   G4   A4
  draw           Eb3  Gb3   G3   A3  Bb3   C4   D4  Eb4  Gb4   G4   A4  Bb4
  blow+slider     D4  Eb4  Gb4   G4   A4  Bb4   C5   D5  Eb5  Gb5   G5   A5
  draw+slider    Eb4  Gb4   G4   A4  Bb4   C5   D5  Eb5  Gb5   G5   A5  Bb5
```

```
World / Raga Bhairav C
  blow            C3  Db3   E3   F3   G3  Ab3   B3   C4  Db4   E4   F4   G4
  draw           Db3   E3   F3   G3  Ab3   B3   C4  Db4   E4   F4   G4  Ab4
  blow+slider     C4  Db4   E4   F4   G4  Ab4   B4   C5  Db5   E5   F5   G5
  draw+slider    Db4   E4   F4   G4  Ab4   B4   C5  Db5   E5   F5   G5  Ab5
```

```
World / Hungarian Minor A
  blow            A2   B2   C3  Eb3   E3   F3  Ab3   A3   B3   C4  Eb4   E4
  draw            B2   C3  Eb3   E3   F3  Ab3   A3   B3   C4  Eb4   E4   F4
  blow+slider     A3   B3   C4  Eb4   E4   F4  Ab4   A4   B4   C5  Eb5   E5
  draw+slider     B3   C4  Eb4   E4   F4  Ab4   A4   B4   C5  Eb5   E5   F5
```

```
World / Hirajoshi E
  blow            E2  Gb2   G2   B2   C3   E3  Gb3   G3   B3   C4   E4  Gb4
  draw           Gb2   G2   B2   C3   E3  Gb3   G3   B3   C4   E4  Gb4   G4
  blow+slider     E3  Gb3   G3   B3   C4   E4  Gb4   G4   B4   C5   E5  Gb5
  draw+slider    Gb3   G3   B3   C4   E4  Gb4   G4   B4   C5   E5  Gb5   G5
```

```
World / Pelog C
  blow            C3  Db3  Eb3   G3  Ab3   C4  Db4  Eb4   G4  Ab4   C5  Db5
  draw           Db3  Eb3   G3  Ab3   C4  Db4  Eb4   G4  Ab4   C5  Db5  Eb5
  blow+slider     C4  Db4  Eb4   G4  Ab4   C5  Db5  Eb5   G5  Ab5   C6  Db6
  draw+slider    Db4  Eb4   G4  Ab4   C5  Db5  Eb5   G5  Ab5   C6  Db6  Eb6
```

```
World / Slendro C
  blow            C3   D3   F3   G3   A3   C4   D4   F4   G4   A4   C5   D5
  draw            D3   F3   G3   A3   C4   D4   F4   G4   A4   C5   D5   F5
  blow+slider     C4   D4   F4   G4   A4   C5   D5   F5   G5   A5   C6   D6
  draw+slider     D4   F4   G4   A4   C5   D5   F5   G5   A5   C6   D6   F6
```

```
World / Double Harmonic E
  blow            E2   F2  Ab2   A2   B2   C3  Eb3   E3   F3  Ab3   A3   B3
  draw            F2  Ab2   A2   B2   C3  Eb3   E3   F3  Ab3   A3   B3   C4
  blow+slider     E3   F3  Ab3   A3   B3   C4  Eb4   E4   F4  Ab4   A4   B4
  draw+slider     F3  Ab3   A3   B3   C4  Eb4   E4   F4  Ab4   A4   B4   C5
```

```
Frasil / Glacier C
  blow            C2   E2   G2   B2   C3   E3   G3   B3   C4   E4   G4   B4
  draw            D2   F2   A2   D3   F3   A3   D4   F4   A4   D5   F5   A5
  blow+slider     C3   E3   G3   B3   C4   E4   G4   B4   C5   E5   G5   B5
  draw+slider     D3   F3   A3   D4   F4   A4   D5   F5   A5   D6   F6   A6
```

```
Frasil / Lydian Haze D
  blow            D2  Gb2   A2  Db3   D3  Gb3   A3  Db4   D4  Gb4   A4  Db5
  draw            E2  Ab2   B2   E3  Ab3   B3   E4  Ab4   B4   E5  Ab5   B5
  blow+slider     D3  Gb3   A3  Db4   D4  Gb4   A4  Db5   D5  Gb5   A5  Db6
  draw+slider     E3  Ab3   B3   E4  Ab4   B4   E5  Ab5   B5   E6  Ab6   B6
```

```
Frasil / Dusk A
  blow            A2   C3   E3   G3   A3   C4   E4   G4   A4   C5   E5   G5
  draw            B2   D3  Gb3   B3   D4  Gb4   B4   D5  Gb5   B5   D6  Gb6
  blow+slider     A3   C4   E4   G4   A4   C5   E5   G5   A5   C6   E6   G6
  draw+slider     B3   D4  Gb4   B4   D5  Gb5   B5   D6  Gb6   B6   D7  Gb7
```

```
Frasil / Aeolian Tide F
  blow            F2  Ab2   C3  Eb3   F3  Ab3   C4  Eb4   F4  Ab4   C5  Eb5
  draw            G2  Bb2  Eb3   G3  Bb3  Eb4   G4  Bb4  Eb5   G5  Bb5  Eb6
  blow+slider     F3  Ab3   C4  Eb4   F4  Ab4   C5  Eb5   F5  Ab5   C6  Eb6
  draw+slider     G3  Bb3  Eb4   G4  Bb4  Eb5   G5  Bb5  Eb6   G6  Bb6  Eb7
```

```
Frasil / Snow Pentatonic G
  blow            G2   B2   D3   E3   G3   B3   D4   E4   G4   B4   D5   E5
  draw            A2   D3   E3   A3   D4   E4   A4   D5   E5   A5   D6   E6
  blow+slider     G3   B3   D4   E4   G4   B4   D5   E5   G5   B5   D6   E6
  draw+slider     A3   D4   E4   A4   D5   E5   A5   D6   E6   A6   D7   E7
```

```
Frasil / Open Fifths E
  blow            E2  Gb2   B2  Db3   E3  Gb3   B3  Db4   E4  Gb4   B4  Db5
  draw            G2   A2   D3   G3   A3   D4   G4   A4   D5   G5   A5   D6
  blow+slider     E3  Gb3   B3  Db4   E4  Gb4   B4  Db5   E5  Gb5   B5  Db6
  draw+slider     G3   A3   D4   G4   A4   D5   G5   A5   D6   G6   A6   D7
```

```
Frasil / Mixolydian Eb
  blow           Eb2   G2  Bb2   C3  Eb3   G3  Bb3   C4  Eb4   G4  Bb4   C5
  draw            F2  Ab2  Db3   F3  Ab3  Db4   F4  Ab4  Db5   F5  Ab5  Db6
  blow+slider    Eb3   G3  Bb3   C4  Eb4   G4  Bb4   C5  Eb5   G5  Bb5   C6
  draw+slider     F3  Ab3  Db4   F4  Ab4  Db5   F5  Ab5  Db6   F6  Ab6  Db7
```

```
Frasil / Harmonics D
  blow            D2   A2   D3  Gb3   A3   C4   D4   E4  Gb4  Ab4   A4  Bb4
  draw            A2  Gb3   C4   E4  Ab4  Bb4  Db5  Eb5   F5   G5  Ab5  Bb5
  blow+slider     D3   A3   D4  Gb4   A4   C5   D5   E5  Gb5  Ab5   A5  Bb5
  draw+slider     A3  Gb4   C5   E5  Ab5  Bb5  Db6  Eb6   F6   G6  Ab6  Bb6
```

```
Order test
  blow            C3  Db3   D3  Eb3   E3   F3  Gb3   G3  Ab3   A3  Bb3   B3
  draw            C4  Db4   D4  Eb4   E4   F4  Gb4   G4  Ab4   A4  Bb4   B4
  blow+slider     C5  Db5   D5  Eb5   E5   F5  Gb5   G5  Ab5   A5  Bb5   B5
  draw+slider     C6  Db6   D6  Eb6   E6   F6  Gb6   G6  Ab6   A6  Bb6   B6
```